

PART III : MATHEMATICS

SECTION – I (Total Marks : 24)

(Single Correct Answer Type)

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

41. Let $P(6,3)$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal at the point P intersects the x -axis at $(9, 0)$, then the eccentricity of the hyperbola is

- (A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\sqrt{2}$ (D) $\sqrt{3}$

ANSWER : B

42. A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0,$$

have one root in common is

- (A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

ANSWER : B

43. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form

$$\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix},$$

where each of a , b , and c is either ω or ω^2 . Then the number of distinct matrices in the set S is

- (A) 2 (B) 6 (C) 4 (D) 8

ANSWER : A

44. The circle passing through the point $(-1, 0)$ and touching the y -axis at $(0, 2)$ also passes through the point

- (A) $\left(-\frac{3}{2}, 0\right)$ (B) $\left(-\frac{5}{2}, 2\right)$ (C) $\left(-\frac{3}{2}, \frac{5}{2}\right)$ (D) $(-4, 0)$

ANSWER : D

45. If

$$\lim_{x \rightarrow 0} \left[1 + x \ln(1 + b^2) \right]^{\frac{1}{x}} = 2b \sin^2 \theta, \quad b > 0 \text{ and } \theta \in (-\pi, \pi],$$

then the value of θ is

- (A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$ (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$

ANSWER : D

46. Let $f : [-1, 2] \rightarrow [0, \infty)$ be a continuous function such that $f(x) = f(1-x)$ for all $x \in [-1, 2]$.

Let $R_1 = \int_{-1}^2 x f(x) dx$, and R_2 be the area of the region bounded by $y = f(x)$, $x = -1$, $x = 2$, and the x -axis. Then

- (A) $R_1 = 2R_2$ (B) $R_1 = 3R_2$ (C) $2R_1 = R_2$ (D) $3R_1 = R_2$

ANSWER : C

47. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is

- (A) $\pm \sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$ (B) $\pm \sqrt{n\pi}, n \in \{1, 2, \dots\}$
 (C) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$ (D) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

ANSWER : A

48. Let (x, y) be any point on the parabola $y^2 = 4x$. Let P be the point that divides the line segment from $(0, 0)$ to (x, y) in the ratio 1:3. Then the locus of P is

- (A) $x^2 = y$ (B) $y^2 = 2x$ (C) $y^2 = x$ (D) $x^2 = 2y$

ANSWER : C

SECTION – II (Total Marks : 16)

(Multiple Correct Answer(s) Type)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** may be correct.

49. If

$$f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \leq -\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \leq 0 \\ x - 1, & 0 < x \leq 1 \\ \ln x, & x > 1, \end{cases}$$

then

- (A) $f(x)$ is continuous at $x = -\frac{\pi}{2}$ (B) $f(x)$ is not differentiable at $x = 0$
 (C) $f(x)$ is differentiable at $x = 1$ (D) $f(x)$ is differentiable at $x = -\frac{3}{2}$

ANSWER : ABCD

50. Let E and F be two independent events. The probability that exactly one of them occurs is $\frac{11}{25}$ and the probability of none of them occurring is $\frac{2}{25}$. If $P(T)$ denotes the probability of occurrence of the event T , then

- (A) $P(E) = \frac{4}{5}, P(F) = \frac{3}{5}$ (B) $P(E) = \frac{1}{5}, P(F) = \frac{2}{5}$
 (C) $P(E) = \frac{2}{5}, P(F) = \frac{1}{5}$ (D) $P(E) = \frac{3}{5}, P(F) = \frac{4}{5}$

ANSWER : AD

51. Let L be a normal to the parabola $y^2 = 4x$. If L passes through the point $(9, 6)$, then L is given by

- (A) $y - x + 3 = 0$ (B) $y + 3x - 33 = 0$
 (C) $y + x - 15 = 0$ (D) $y - 2x + 12 = 0$

ANSWER : ABD

52. Let $f : (0,1) \rightarrow \mathbb{R}$ be defined by

$$f(x) = \frac{b-x}{1-bx},$$

where b is a constant such that $0 < b < 1$. Then

- (A) f is not invertible on $(0, 1)$
 (B) $f \neq f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$
 (C) $f = f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$
 (D) f^{-1} is differentiable on $(0, 1)$

ANSWER : A

SECTION – III (Total Marks : 24)

(Integer Answer Type)

This section contains **6 questions**. The answer to each of the questions is a **single-digit integer**, ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS.

53. Let $\omega = e^{i\pi/3}$, and a, b, c, x, y, z be non-zero complex numbers such that

$$a + b + c = x$$

$$a + b\omega + c\omega^2 = y$$

$$a + b\omega^2 + c\omega = z.$$

Then the value of $\frac{|x|^2 + |y|^2 + |z|^2}{|a|^2 + |b|^2 + |c|^2}$ is

ANSWER : MARKS TO ALL

54. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is

ANSWER : 2

55. Let $y'(x) + y(x)g'(x) = g(x)g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$, where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given non-constant differentiable function on $\mathbb{R}$ with $g(0) = g(2) = 0$. Then the value of $y(2)$ is

ANSWER : 0

56. Let M be a 3×3 matrix satisfying

$$M \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, \quad M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad \text{and} \quad M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}.$$

Then the sum of the diagonal entries of M is

ANSWER : 9

57. Let $\vec{a} = -\hat{i} - \hat{k}$, $\vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then the value of $\vec{r} \cdot \vec{b}$ is

ANSWER : 9

58. The straight line $2x - 3y = 1$ divides the circular region $x^2 + y^2 \leq 6$ into two parts. If

$$S = \left\{ \left(2, \frac{3}{4} \right), \left(\frac{5}{2}, \frac{3}{4} \right), \left(\frac{1}{4}, -\frac{1}{4} \right), \left(\frac{1}{8}, \frac{1}{4} \right) \right\},$$

then the number of point(s) in S lying inside the smaller part is

ANSWER : 2

SECTION – IV (Total Marks : 16)

(Matrix-Match Type)

This section contains **2 questions**. Each question has **four statements** (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.

59. Match the statements given in **Column I** with the values given in **Column II**

Column I	Column II
(A) If $\vec{a} = \hat{j} + \sqrt{3}\hat{k}$, $\vec{b} = -\hat{j} + \sqrt{3}\hat{k}$ and $\vec{c} = 2\sqrt{3}\hat{k}$ form a triangle, then the internal angle of the triangle between $\vec{a}$ and $\vec{b}$ is	(p) $\frac{\pi}{6}$
(B) If $\int_a^b (f(x) - 3x)dx = a^2 - b^2$, then the value of $f\left(\frac{\pi}{6}\right)$ is	(q) $\frac{2\pi}{3}$
(C) The value of $\frac{\pi^2}{\ln 3} \int_{\frac{1}{6}}^{\frac{5}{6}} \sec(\pi x) dx$ is	(r) $\frac{\pi}{3}$
(D) The maximum value of $\left \operatorname{Arg} \left(\frac{1}{1-z} \right) \right $ for $ z =1$, $z \neq 1$ is given by	(s) π
	(t) $\frac{\pi}{2}$

ANSWER A : q

B : p or p, q, r, s and t

C : s

D : t

60. Match the statements given in **Column I** with the intervals/union of intervals given in **Column II**

Column I

- (A) The set $\left\{ \operatorname{Re}\left(\frac{2iz}{1-z^2}\right) : z \text{ is a complex number, } |z|=1, z \neq \pm 1 \right\}$ is
- (B) The domain of the function $f(x) = \sin^{-1}\left(\frac{8(3)^{x-2}}{1-3^{2(x-1)}}\right)$ is
- (C) If $f(\theta) = \begin{vmatrix} 1 & \tan \theta & 1 \\ -\tan \theta & 1 & \tan \theta \\ -1 & -\tan \theta & 1 \end{vmatrix}$, then the set $\left\{ f(\theta) : 0 \leq \theta < \frac{\pi}{2} \right\}$ is
- (D) If $f(x) = x^{3/2}(3x-10)$, $x \geq 0$, then $f(x)$ is increasing in

Column II

- (p) $(-\infty, -1) \cup (1, \infty)$
- (q) $(-\infty, 0) \cup (0, \infty)$
- (r) $[2, \infty)$
- (s) $(-\infty, -1] \cup [1, \infty)$
- (t) $(-\infty, 0] \cup [2, \infty)$

ANSWER A : s

B : t

C : r

D : r